Overcoming challenges in crash prediction modeling using discretized duration approach: An investigation of sampling approaches

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Abstract

Until recently, statistical approaches used for real-time crash prediction modeling were limited to case-control design and "sampling of alternatives" approaches. A recent study has developed a duration-based real-time crash prediction model capable of incorporating dynamic (time-varying) covariates within its framework. The modeling approach discretizes the duration between crashes into equal time intervals which can be modeled as alternatives in a multinomial logit framework. The approach, however, requires a reformulation of the original crash dataset to fit its modeling framework which results in considerably large data making model estimation computationally demanding. Additionally, validation of the model in the original study is based on crash data from just one interstate, I-405, assuming homogenous highway segments each 5 miles in length. This study improves upon the original study by investigating sampling techniques that can be applied to the reformulated data to reduce computational load using 2019 crash data from two interstates, I-40 and I-55, in Memphis, Tennessee. Furthermore, discretization of inter-crash duration is undertaken following non-homogenous segmentation of the interstates that is based on highway geometry, terrain, and posted speed limit. To accomplish the study objectives, a relatively small future window of 1 hour with 15-minute time intervals is used to discretize the inter-crash duration and obtain the reformulated data. Sampling of crashes for model estimation is then done at the crash, epoch, and segment levels to answer the question of which sampling technique (by crash, epoch, or segment) would result in reasonable accuracy when compared with the complete (100%) data. Results show that 25% of samples drawn at the epoch level can result in a considerable reduction of computational load while providing reasonably consistent estimates.

Keywords: real-time crash prediction; crash data sampling; discrete choice; discretized duration; hazard model; multinomial logit

1. Introduction

Traffic engineers utilize Safety Performance Functions (SPFs) to identify the causal factors of crashes and develop strategies to make highways safer. SPFs are developed using regression techniques and can be of two types: i) static risk models and ii) dynamic risk models depending on the type of data they are trained on (Yasmin et al., 2018). Static risk models regress the frequency of crashes at a particular location as a function of predictors such as traffic flow, highway geometry, environmental conditions, driver, and vehicle characteristics (e.g., age, gender, vehicle body type, etc.) using aggregated traffic and crash data collected from the location over an extended period (e.g., see (Imprialou et al., 2016; Ma et al., 2008; Ma & Kockelman, 2006)). Although useful for safety interventions, there are two main limitations to static risk models. First, aggregated data used to train these models fail to capture variations in the predictor variable which can result in erroneous inferences of the relationship between crash and its predictor. For example, traffic flow can vary considerably across a day, month, and year. Aggregated metrics such as Average Annual Daily Traffic, when used as a precursor to crashes, cannot capture the variation in traffic flow conditions that influences crashes. Second, static risk models are only useful for reactive traffic management, it lends little to no utility towards proactive traffic management. This necessitates the use of dynamic risk models.

Dynamic risk models or real-time crash prediction models use real-time disaggregated traffic data to eliminate the limitations of static risk models. Contrary to its static counterpart, dynamic risk models evaluate the probability of crash based on observed associations between traffic and crash occurrences, thus enabling unsafe traffic conditions to be used as predictors of future crashes (Hossain et al., 2019; Lee et al., 2003; Theofilatos et al., 2019). This enables traffic engineers and planners to monitor real-time traffic flow dynamics, identify unsafe conditions, make predictions, and take proactive measures in real-time to fix traffic flow conditions and avoid unfavorable outcomes such as crashes and congestion.

With the advent of advanced data-driven techniques such as data mining and Machine Learning (ML), researchers no longer need to depend on regression techniques to develop SPFs, and rightfully so. ML algorithms have been found to offer better predictive accuracy compared to traditional SPFs (Ariannezhad et al., 2021; Cai et al., 2020). However, data-driven techniques are limited by their transportability (i.e., the ability to generalize causal relationships in scenarios other than what the algorithm is trained on) and ability to provide insight into causal inferences (Mannering et al., 2020). Therefore, SPFs are still desired when the extraction of causal inferences is of interest. In particular, regression equations grant the ability to draw interpretable results through variable coefficients and marginal effects which is desired when generating crash forecasts under different policy scenarios.

2. Literature review

2.1. Statistical approaches in real-time crash modeling

Regression-based real-time crash prediction models in the literature are largely based on a case-control approach where crashes are matched with non-crash events and crash likelihood is modeled using the binary logit framework. For example, crashes can be considered to have binary outcomes (crash=1, no crash=0) but crash events are seldomly encountered compared to non-crash events. Therefore, every crash is matched with multiple non-crash events based on common variables such as location, time and day, real-time traffic flow, etc. which serve as controls (M. Abdel-Aty & Pande, 2007; Rahman et al., 2021; Sun & Sun, 2016). However, an issue that arises with this approach is the ratio of events to its controls usually is not representative of the actual phenomenon and the optimal ratio often varies from one study to another (Yasmin et al., 2018). Recently, a “sampling of alternatives” approach was proposed by researchers to overcome this shortcoming (Yasmin et al., 2018). In this approach, the analysis window (1 month) was divided into 5-minute intervals. A crash could occur on any of these time intervals, therefore allowing the time intervals to be modeled as alternatives in a Multinomial Logit (MNL) model. To avoid computational

issues arising from a large number of alternatives, the researchers used a sampling approach where 30 alternatives were drawn which included the time interval where the crash was observed and 29 other random intervals. Using this approach, the authors developed a joint model by integrating monthly crash risk and real-time crash risk that could predict crashes for 5-minute time intervals in the next month. While it is appealing to forecast potential crashes far into the future, a future window of 1 month considered in the study is impractical. There can be considerable changes in traffic flow and weather conditions over longer periods which makes predictions with longer windows unrealistic and unreliable.

Another notable approach in the prediction of crash likelihood is the use of duration models, also called hazard or survival models. These models describe the conditional probability of an event (in this case a crash) occurring at a certain time *t* provided it has not occurred until then. Hazard models in their most basic form assume a constant effect of a predictor variable on the outcome over time which is referred to as constant hazard rate. The framework was first implemented to analyze crash data in 1990 (H. L. Chang & Jovanis, 1990). It has since been extended to model crashes on highway intersections (e.g., (Bagloee & Asadi, 2016)**)**, predict clearance time for roadway incidents (e.g., (Nam & Mannering, 2000)), predict crashes on highway work zones (e.g., (Thapa & Mishra, 2021)**)**. Its application on real-time crash prediction, however, is not straightforward because of its inability to incorporate time-varying covariates. However, if the duration between crashes is divided into equal-time intervals and modeled as outcomes in a choice model, time-varying covariates can be introduced in duration models as interval-specific variables (Paleti et al., 2021).

TABLE 1. Types of sampling techniques with examples

Technique	Study	Description
Random	(Zimmerman et al., 2012)*	Random selection of GPS points from a geographical cluster within a city to select individuals and households to be interviewed on traffic crashes
Interval	(Al-Ghamdi, 2002)	Investigated the determinants of pedestrian crashes by selecting every third record from a list of pedestrian crashes.
Stratified	(Theofilatos et al., 2019)	Case-control approach with samples drawn from non-crash events to estimate the effect of real-time traffic characteristics on crash occurrence
	(Harb et al., 2008)	Stratified sampling based on number of lanes, speed limit and time of day (a.m. or p.m.) to compare work zone and non-work zone crashes (necessitated by varying traffic flow conditions at these crash locations) using conditional logistic regression
	(Mohan et al., 2017)	Compared cities stratified by traffic fatality rates to investigate the effect of road type and junction density on crashes
Cluster	(Furth, 2011; Furth et al., 1988)	Sampling of trips (e.g., of trips-round trips or chain of trips on a single route) to estimate true transit patronage
	(Zimmerman et al., 2012)*	Geographical clustering of the study area within cities based on their GPS coordinates to identify clusters of individuals and households who could be interviewed on traffic crashes

*Note: (Zimmerman et al., 2012) use two sampling techniques in their study. The first is geographical clustering to divide cities into grids. Random GPS coordinates were then drawn from the grid to select individuals and households for their survey and interviews.

2.2. Sampling approaches

Developing regression models using large crash data can be computationally demanding. In this regard, probability sampling techniques can help draw representative samples that can provide accurate statistical inferences with a reduced computational load and time. Probability sampling techniques draw samples from

the original data at random, therefore, assigning every observation an equal probability of being selected. Since the current study necessitates the use of sampling techniques, we briefly describe the probability sampling techniques as they are relevant to this study.

Probability sampling can be done in one of four following ways: (i) simple random sampling by drawing samples at random; (ii) systematic or interval sampling by drawing a random sample at first and samples after a certain interval then after (every n^{th} sample); (iii) stratified sampling by drawing a sample from preidentified groups or strata in the data; and (iv) cluster sampling by identifying and selecting natural clusters within the data. The approaches in sampling have their distinct advantages and disadvantages. Simple and systemic approaches are the easiest to perform. However, for imbalanced data (where a particular observation is rare while others are abundant) simple random sampling can lead to biases. Interval sampling can also lead to biases particularly when the data being drawn from has a periodical or repetitive structure such as those observed in panel and cross-sectional data. The approach also requires a complete list of observations before being administered. Stratified and cluster sampling although immune to biases that plague simple random and interval sampling, are particularly complicated to work with. In stratified sampling, having prior knowledge of the data properties is necessary to identify groups and strata in the data. While this can be difficult, it is particularly useful in the case of imbalanced data as samples can be drawn from groups of frequent observations to match rare observations (e.g., draw only from non-crash events to match it with crash events). Although alternatively, synthetic data generation techniques can also be used to create synthetic data points and eliminate the need for sampling (e.g., see (Ariannezhad et al., 2021; Cai et al., 2020; Ke et al., 2019). Similarly, in cluster sampling, it is challenging to determine the number of clusters required to represent large data, and this gives rise to statistical uncertainty. Table 1 describes selected studies based on the sampling techniques. The list of studies in the table is not meant to be an exhaustive one and is provided here to introduce them to the reader.

2.3. Role of traffic flow parameters

Traffic flow parameters are established precursors of highway crashes. Hence, their inclusion in crash prediction models is a requisite. Despite this, their specific influence on crashes has been disparate across studies. Specifically, investigations on the influence of speed on crash occurrence have reported contesting results. Although a direct relationship between speed and crash severity has been reported consistently, the speed-crash frequency relationship remains questionable. The relationship between them varies across the literature from negative (Baruya, 1998), insignificant (Ma et al., 2008; Quddus, 2013), to positive (Kloeden et al., 2002). Concerning the influence of traffic volumes on the speed-crash relationship, researchers have emphasized the inclusion of real-time traffic volumes and vehicle occupancy (Aarts & van Schagen, 2006; Garber & Ehrhart, 2000) since both higher and lower traffic volumes have been reported as determinants of crashes which has been attributed to the speed variance associated with the change in traffic volumes (M. A. Abdel-Aty & Radwan, 2000; L.-Y. Chang, 2005; Garber & Ehrhart, 2000).

3. Research objectives

As mentioned before, the use of duration models in real-time crash prediction models is non-existent except for (Paleti et al., 2021). In the study, the authors demonstrate their conceptual framework using crash data from I-405 assuming homogenous 5-mile highway segments. Their study uses a five-hour epoch with one-hour time intervals as the future window for crash prediction. The current study leverages potential scopes for improving the original paper while also further validating the model.

The assumption of homogenous segments, that is, traffic and highway geometry do not vary across a link, is done to simplify the modeling approach. In reality, traffic and highway geometry vary significantly within links (Imprialou et al., 2016). Therefore, additional analysis with appropriate highway segmentation and multiple interstates is necessitated to further validate and showcase the framework's applicability.

The framework requires a reformulation of the crash data (i.e., discretization of inter-crash duration into small time intervals) to fit into its framework. This results in a considerable increase in data size (please refer to the *Model Framework* section for a detailed description of data reformulation and how it increases in data size). The issues of data size become even more pronounced when a smaller time discretization is used. While the authors have validated the model using a future window of five hours with 1-hour intervals, a smaller time discretization is more practical. Therefore, the issue of data size becomes inevitable and needs to be addressed.

To address these limitations, current research aims to answer the question, "*If the discretized duration-based crash modeling framework is to be used, utilization of the complete reformulated dataset becomes too exhaustive and computationally demanding. In such a case, what sampling technique(s) and sample size(s) provides the most time-efficient and consistent estimates?".* In doing so, this research further validates and improves upon the original framework by i) utilizing crash data from multiple interstates segmented by geometry (number of lanes), terrain (flat, rolling or mountainous), and traffic flow (posted speed limit), ii) using a smaller and more realistic future window of 1 hour further discretized into 15-minute intervals for prediction, and iii) investigating sampling techniques and sample size required at different sampling levels (more on this in the *Modeling Approach* section) to derive consistent estimates while also reducing the computational complexity arising from large data size.

The rest of the paper is outlined as follows. We present a detailed description of the framework and the data reformulation process from the original paper in the *Model framework* section. The study area and data used in the study are then discussed in the *Data* section. The section: *Sampling approach* expands on the approach taken in sampling crash data and estimation models. The *Results* section discusses the results from our models followed by the *Discussion* section where we examine our findings and its implications. The *Conclusion* section summarizes the findings from this study and presents avenues for future research.

4. Model framework

The hazard function *h(t)* in a duration-based crash model gives the probability of a crash occurring at a certain time *t* conditional upon no crashes having occurred until then. Assuming a constant hazard rate over time (*h*), the hazard function can be written as.

$$h(t) = \frac{f(t)}{1 - F(t)} = \frac{he^{-ht}}{1 - (1 - e^{-ht})} = h \tag{1}$$

In equation 1, *f(t)* is the probability distribution function and *F(t)* is the probability density function for a continuous random variable *T*. As such, *F(t)* gives the probability that a crash has occurred by *t* and it can be expressed using equation 2.

$$F(t) = Pr(T \leq t) \tag{2}$$

Assuming time to be discretized into *n* intervals each with a duration of *dt*, the probability of a crash occurring at a certain interval *n* since the beginning can be written and simplified as follows.

$$\begin{aligned} Pr(T = ndt) &= Pr(T \leq ndt) - Pr(T \leq (n-1)dt) \\ &= F(ndt) - F\big((n-1)dt\big) \\ &= exp\,(-h(n-1)dt) - exp\,(-hndt) \\ &= \frac{exp\,(-h(n-1)dt)}{1/(1 - exp\,(-hdt))} \end{aligned} \tag{3}$$

The denominator in equation 3 can be written using a Taylor series, i.e., $\frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots \infty$ for -1 < *x* < 1 to obtain equation 4.

$$Pr(T = ndt) = \frac{exp(-h(n-1)dt)}{1 + exp(-hdt) + exp(-2hdt) + exp(-3hdt) + \cdots \infty} \tag{4}$$

$$= \frac{exp(U_n)}{exp(U_1) + exp(U_2) + exp(U_3) + \cdots \infty}, \text{where } U_n = -h(n-1)dt$$

$$= \frac{exp\ (U_n)}{\sum_1^{\infty} exp\ (U_c)}$$

After simplification of equation 4, we see that the occurrence of the crash over the nth interval takes the form of a Multinomial Logit (MNL) model with infinite alternatives for n (i.e., $n \rightarrow \infty$). The latent propensity associated with each alternative (U_n) is associated with the hazard rate (h), number of alternatives (n), and time interval (dt) as $U_n = -h(n-1)dt$. This relationship allows the introduction of time-varying covariates and non-linear hazard rates into an MNL framework through the latent propensity function which is a linear combination of various degree polynomials shown in equation 5.

$$U_n = \beta_1(n-1)dt + \beta_2[(n-1)dt]^2 + \beta_2[(n-1)dt]^3 + \cdots \tag{5}$$

When higher-order polynomial terms are zero, the MNL becomes a simple exponential model. The methodological framework used to modify crash data to the discretized modeling framework is discussed in the preceding paragraphs.

For empirical application in this study, we assume that each epoch is one hour, and it is further discretized into four 15-minute intervals, i.e., dt =0.25 hours. We use the indexes e and s to represent epochs and segments henceforth. Therefore, the number of intervals in each epoch is 4, which we represent using the index, C, i.e., C=4. The number of epochs is dependent on the time interval between two consecutive crashes in a segment. For instance, let us assume for roadway segment 1, a second crash is observed 3.5 hours after the first. The inter-crash duration of 3.5 hours is discretized into four epochs wherein the crash occurs at the second 15-minute interval of the fourth epoch as shown in Table 2. Additionally, the column "Beyond 1 hour of the current epoch" provides information on whether the second crash occurred in a certain epoch or the future. In the presented example, the value 1 in the column for the first three epochs communicates that the second crash occurred in the future. Conversely, its value is 0 for the fourth epoch suggesting the crash occurred in the current (fourth) epoch.

If i be the index representing the 15-minute intervals. Then, the time since the beginning of interval i in epoch e can be written as $t_{e,i} = (e-1)Cdt + (i-1)dt$. For the example presented in Table 2, the time for the second interval of the fourth epoch is therefore $t_{4,2} = (4-1)1 + (2-1)0.25 = 3.25$. The latent propensity function for each time interval, i = (1,2,3,4) of an epoch can then be written using equation 6.

TABLE 2. Reformulation of crash data to create forecasting epochs

Segment	Time until next crash (hours)	Epoch	First 15-min	Second 15-min	Third 15-min	Fourth 15-min	Beyond 1 hour of the current epoch
1	3.5	1	0	0	0	0	1
1	3.5	2	0	0	0	0	1
1	3.5	3	0	0	0	0	1
1	3.5	4	0	1	0	0	0

$$U_{s,e,i} = \beta_1 t_{e,i} + \beta_2[t_{e,i}]^2 + \cdots + r'\ X_{s,e,i} \tag{6}$$

As shown by equation 6, each time interval can be considered an alternative in the MNL model. The first two terms on the right-hand side of the latent propensity function introduces the effect of duration dynamics. The final term adds the effect of time-varying covariates into the propensity function. In addition to the four alternatives, there is one additional C+1th alternative that indicates whether a crash occurred in the future (last column in Table 2). The propensity function for this alternative can be written as.

$$U_{s,e,C+1} = \beta_{C+1} \tag{7}$$

The conditional probability of the next crash occurring in time-interval i of epoch e for a segment s and random variable of time T_s, provided no crash has occurred until the previous epoch, can be written using equation 8.

$$Pr\left(T_s = t_{e,i} \middle| T_s > (e-1)Cdt\right) = \frac{exp\ (U_{s,e,i})}{\sum_{c=1}^{C} exp\ (U_{s,e,c}) + exp\ (U_{s,e,C+1})} \tag{8}$$

The unconditional probability of a crash occurring at interval i of epoch e can then be calculated using equation 9.

$$Pr(T_s = t_{e,i}) = \frac{exp\ (U_{s,e,i})}{\sum_{c=1}^{C} exp\ (U_{s,e,c}) + exp\ (U_{s,e,C+1})} \cdot \prod_{e*=1}^{e-1} \frac{exp\ (U_{s,e*,C+1})}{\sum_{c=1}^{C} exp\ (U_{s,e*,c}) + exp\ (U_{s,e*,C+1})} \tag{9}$$

The parameters in the MNL model can be estimated by maximizing the product of the unconditional likelihood function in Equation 9. This is done by defining a vector of parameters,$n = (\beta_1, \beta_2, \dots, r, \beta_{C+1})'$. The vector is obtained by vertical concatenation of parameters in the latent propensity function.

5. Data

One of the objectives of the study was to include multiple interstates in model estimation. Therefore, crashes from two primary interstates within Memphis city limits in Tennessee were used for the study. Crash data for the year 2019 was collected from Enhanced Tennessee Roadway Information Management System (ETRIMS) and dynamic data related to traffic flow parameters were obtained from Radar Detection System (RDS) stations. Since RDS stations in Tennessee are mostly located within city boundaries, only those crashes that occurred on primary interstates, I-40 and I-55, within Memphis were chosen. Our analysis included crashes on 21.51 miles of I-40 and 12.28 miles of I-55. Figure 1 shows the location of the city and the two interstates superimposed over the state map of Tennessee. Discretized crash data for analysis was obtained from the study area as follows.

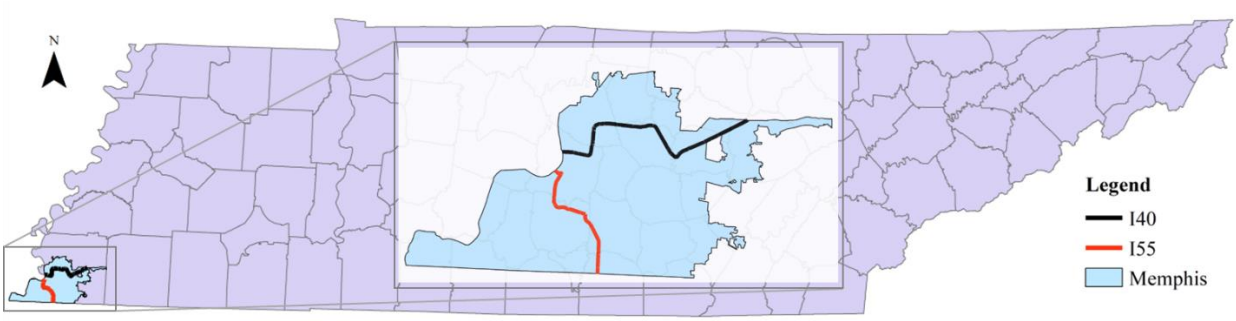


Figure 1. Study area.

ETRIMS, Tennessee Department of Transportation's main portal for transportation-related data, was used to extract crash data. ETRIMS is an online query-based system that stores information on highway geometry, aggregated traffic flow, and highway crashes in the state. The highway inventory information in ETRIMS formed the basis for our highway segmentation. Specifically, highway segmentation was based on i) the direction of traffic, ii) the number of lanes, iii) terrain (flat or rolling), and iv) posted speed limit. This produced 146 segments for I-40, and 94 segments for I-55. Crash data for the study area also included exact coordinates which enabled mapping each crash to its respective segment based on geographical

proximity and direction of travel. For each segment, the duration between consecutive crashes was then determined using the date and time of the crash. The inter-crash duration was discretized into 1-hour epochs with 15-min intervals as presented in Table 2. To avoid confusion, it is necessary to distinguish between crash data retrieved from ETRIMS and crash data obtained after the creation of forecasting epochs. Therefore, we will refer to the process of creating forecasting epochs as data expansion and the data obtained from it as expanded data. The crash data from ETRIMS (without forecasting epochs) will be referred to as original crash data.

It is also worth noting here that our study excluded crashes occurring near entrance and exit ramps since traffic flow parameters vary significantly in these locations. This resulted in a total of 1,174 crashes on the two interstates. Table 3 presents the descriptive statistics of variables in the study. Figure 2 shows the distribution of inter-crash duration for all the interstate segments. The distribution of inter-crash duration suggests that about 62% of crashes occurred within 500 hours (~21 days) of the first crash. Notably, the mean and median durations were 652 hours and 324 hours respectively suggesting a right-skewed distribution. The minimum duration between crashes was 0 hours, suggesting at least two different crashes in a segment occurred at the same time. The maximum duration between crashes was 6,608 hours.

TABLE 3. Descriptive statistics of variables

Categorical variables	Frequency	Percentage
Time of day		
Early morning (6 a.m. to 9 a.m.)	208	17.72
Late morning (9 a.m. to 12 p.m.)	132	11.24
Early afternoon (12 p.m. to 3 p.m.)	175	14.91
Late afternoon (3 p.m. to 6 p.m.)	292	24.87
Evening (6 p.m. to 12 a.m.)	208	17.72
Night (12 a.m. to 6 a.m.)	159	13.54
Weather condition		
Clear	849	72.32
Cloudy, rain, fog, or snow	325	27.68
Lighting condition		
Daylight	758	64.57
Dark lighted	287	24.45
Dark, not lighted	129	10.99
Terrain		
Flat	640	54.51
Rolling	534	45.49
Land use		
Rural	448	38.16
Commercial	595	50.68
Mixed	131	11.16

Continuous variables	Min	Q1	Median	Q3	Max	Mean	SD
Highway geometry							
Number of lanes (one-direction)	2	3	4	5	6	3.89	1.07
Inter-crash duration (hours)	0	105	324	865.50	6,608	652	863.75

ETRIMS stores aggregated data on traffic flow parameters, therefore, dynamic traffic flow for the study area was collected from a different source. Specifically, RDS stations spread out across the interstate segments were used to obtain dynamic or time-varying traffic flow parameters for each crash. First, RDS stations were mapped to their respective highway segments (and therefore crashes) based on geographical proximity and direction of travel. Three traffic flow parameters, viz, speed, volume, and occupancy sampled every 15-min were obtained for each RDS station mapped to the crashes for the year 2019. The sampled flow parameters were then matched against the discretized 15-min time intervals between crashes using timestamps. The resulting database, therefore, included static and dynamic co-variates for every 15-min interval of an epoch for all crashes in the study area.

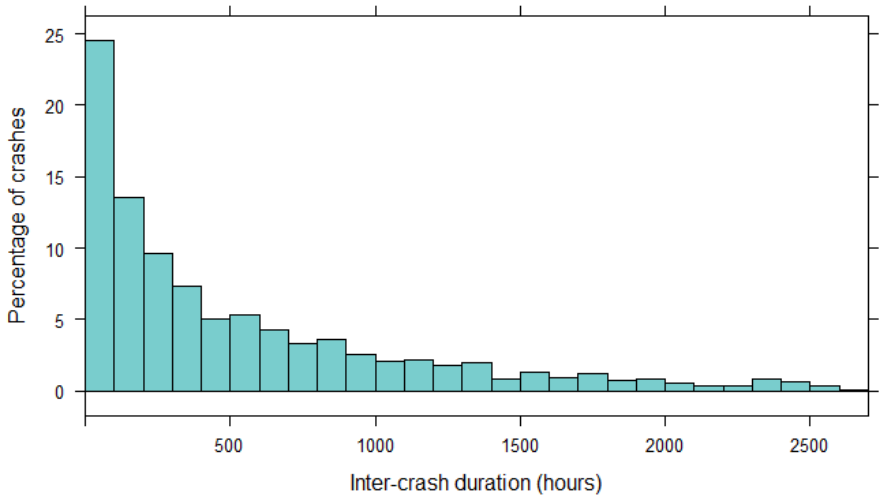


Figure 2. Distribution of inter-crash duration

6. Sampling approach

As evident from our discussion of the methodology, the size of the expanded data depends on three factors, i) the size of original crash data, ii) inter-crash duration between crashes, and iii) choice of time discretization. With an increase in these factors, the expanded data enlarges which brings forth computational challenges. To overcome this, random sampling techniques with progressive sampling can be employed to identify the smallest sample that can best represent the complete dataset (Provost et al., 2001). In this method, the size of random samples used for training is gradually increased while checking whether the model accuracy increases with an increase in sample size. The smallest sample size can then be chosen for subsequent modeling. For this study, we follow a similar approach wherein different sizes of the training data are randomly sampled at different levels. Models trained on these samples are then compared against models trained on the complete training data and tested on training data. It is worth pointing out that this is practical for our dataset due to the relatively small region and time frame chosen for analysis. Our findings can however be generalized and incorporated in future studies that apply the current modeling framework to analyze crashes for a larger region and time frame. Next, we illustrate the approach used sampling data for training and testing the duration-based MNL model.

To compare model performance for various samples, it is necessary to train the MNL on different samples but test on the same. Therefore, while models were trained on samples derived from various sampling approaches, they were all tested on the same testing data. Specifically, 90% of original crash data was selected for training and the remaining 10% was held out for testing. All subsequent samples for model training were derived from the training data. Similarly, the expanded test data was used to test all trained models. Based on different sampling approaches, ten models were trained and tested on the expanded

testing data. The samples used to train these models were drawn at three levels, crash, epoch, and segments as summarized in Table 4.

6.1. Crash level sampling

At the crash level, a specific percentage of crashes were drawn from the original training set and then expanded. Specifically, we drew 5%, 25%, 50%, and 100% of original crashes and expanded them to form the training data for the crash level sampling. This was straightforward considering each crash had a unique identifier. In other words, we employed a simple random sampling approach here. These samples drawn at this level are denoted by the prefix *Cr_* in Table 4 and henceforth. For example, *Cr_100%* represents the complete expanded dataset containing all original crashes from the training set.

6.2. Epoch level sampling

For sampling at the epoch level, the original testing crash data was first expanded. Then a specific percentage of observations was drawn at random from the expanded data. It is worth mentioning here that observations in the expanded dataset correspond to various epochs of crashes and not unique crashes. We utilized unique crash identifiers and corresponding epoch numbers to draw unique samples. The number of samples drawn was 5%, 25%, and 50% of the expanded dataset. We identify these samples using the prefix *Ep_*.

6.3. Segment level sampling

To draw samples at the segment level, each roadway segment was assumed to be a stratum. Crashes were then sampled from each stratum. This sampling was carried out using the original training crash data and then expanded. For example, when drawing 5% of crashes at the segment level, 5% of total crashes were drawn such that crashes sampled were from different segments. As with sampling at crash and epoch levels, 5%, 25%, and 50% crashes were sampled based on highway segments. The sampled crash data was then expanded. The resulting expanded dataset is denoted by the prefix *Sg_*.

TABLE 4. Description of training and testing data

Data	Sampling level	Model name	Description
Training	Crash	Cr_100%	100% of the expanded dataset with 100% of original training crash data
		Cr_5%	100% of the expanded dataset with 5% of original training crash data
		Cr_25%	100% of the expanded dataset with 25% of original training crash data
		Cr_50%	100% of the expanded dataset with 50% of original training crash data
Training	Epoch	Ep_5%	5% of the expanded dataset with 100% of original training crash data
		Ep_25%	25% of the expanded dataset with 100% of original training crash data
		Ep_50%	50% of the expanded dataset with 100% of original training crash data
Training	Segment	Sg_5%	100% of the expanded dataset with 5% of original training crashes drawn from segment strata
		Sg_25%	100% of the expanded dataset with 25% of original training crashes drawn from segment strata
		Sg_50%	100% of the expanded dataset with 50% of original training crashes drawn from segment strata
Testing	Crash	-	100% of the expanded dataset with 100% of original testing crash data

Note: Original training crash data = 90% of original crash data, original testing crash data = 10% of original crash data.

7. Results

An initial analysis of variables was conducted to avoid multicollinearity issues in the predictor variables. Results showed a high correlation between occupancy and speed derived from RDS stations. This is expected since higher speed and occupancy are interrelated. Therefore, occupancy was removed from all our subsequent models. Analysis of dynamic traffic volumes obtained from RDS stations showed large variation in data. The volumes were therefore scaled between 0 and 1. No more data, cleaning was needed after this and the MNL was run using complete training data, to obtaing the *Cr_100%* model. Table 5 presents the results from the model trained on *Cr_100%*. In our model estimation, the last alternative (whether the crash occurred beyond the current epoch) was set as the base alternative for which only a constant was estimated irrespective of the segment and epoch. Based on the results, the propensity equation for the base alternative can therefore be written as:

$$U_{s,e,Beyond\ 1\ hour\ of\ current\ epoch}(base) = 5.336$$

Relative to the base case, the propensity equation for the four intervals (*i*) of each epoch (*e*) and segment (*s*) can be written as:

$$U_{s,e,i} = -0.254 \times t_{e,i} + 0.327 \times \left(t_{e,i}\right)^2 - 0.823 \times Early\ morning - 0.723 \times Late\ morning \ldots - 0.029 \times Average\ speed - 0.920 \times Volume$$

TABLE 5. Result for the model trained on *Cr_100%* data.

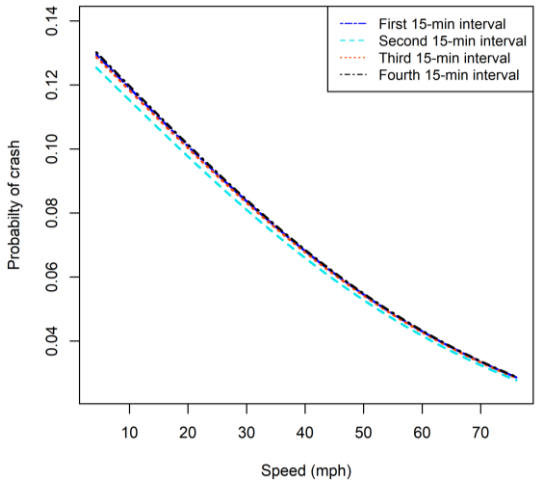
Variable groups	Variables	Parameter	t-stat
Intercept	Intercept	5.336	70.233
Duration dynamics	Time since last crash	-0.254	-44.350
	Square of time since last crash	0.327	41.951
Time of day (base = Night (12 a.m. to 6 a.m.))	Early morning (6 a.m. to 9 a.m.)	-0.823	-63.138
	Late morning (9 a.m. to 12 p.m.)	-0.289	-58.829
	Early afternoon (12 p.m. to 3 p.m.)	-1.005	-73.728
	Late afternoon (3 p.m. to 6 p.m.)	-2.026	-74.878
	Evening (6 p.m. to 12 a.m.)	-0.656	-62.907
Weather condition (base = Clear)	Cloud rain fog or snow	0.024	14.337
Lighting condition (base = Daylight)	Dark lighted	0.147	55.970
	Dark not lighted	0.024	18.659
Terrain (base = Rolling)	Flat	-0.407	-80.277
Land use (base = Mixed)	Rural	-0.376	-89.028
	Commercial	0.097	34.930
Traffic stream characteristics	Average speed	-0.029	-19.336
	15-minute traffic flow	-0.920	-64.593
Highway geometry	Number of lanes	0.060	2.554
Goodness of fit			
Average initial LL		-0.0831	
Average LL at convergence		-0.0139	
Average predicted LL		-0.0161	
Number of observations in the training sample		695,374	
Number of observations in the testing sample		68,544	

Few notable observations can be made from the results in Table 5. First, both time since the last crash and its quadratic polynomial is statistically significant. This suggests a non-linear influence of time

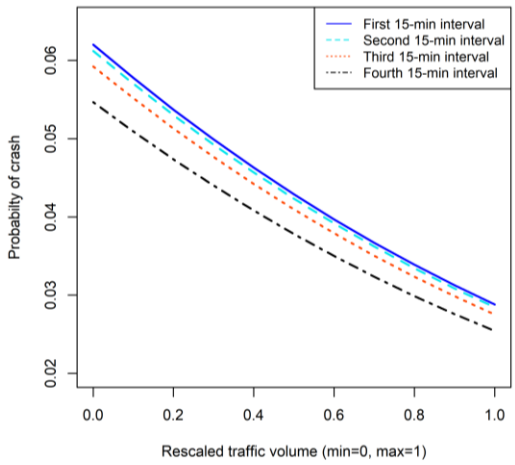
on the occurrence of crashes. Second, compared to night, crashes are the least likely to occur in late afternoons of the epoch compared to future epochs. When the weather condition is unfavorable, there is an increased likelihood of crashes in the current epoch. The same is true for unfavorable lighting conditions. Compared to rolling terrain, in flat terrains crashes are more likely to occur in the future epochs. When looking at the effect of land use, we observed a higher and lower likelihood of crashes in commercial and rural areas respectively. With an increase in vehicle speed and traffic flow, crashes are more likely to occur in the future epochs. This suggests that during better flow conditions, crashes are less likely to occur frequently (i.e., inter-crash duration is higher). However, roadway segments with a larger number of lanes are more likely to encounter crashes during a certain epoch compared to the future.

7.1. Effect of sampling approach and sample sizes on model estimates

Next, we ran the MNL for the remaining nine samples and tested the estimated model using the testing set. The average predicted log-likelihood for the testing set is used to compare the trained models. Table 6 presents the results from our model estimations. We see inconsistent results for models trained on samples drawn at crash and segment levels. Of all the models, models estimated using 25% and 50% of samples drawn at epoch level, *Ep_25%*, and *Ep_50%*, respectively, provided estimates that were closest to *Cr_100%*. The average predicted log-likelihood for the *Ep_25%* and *Ep_50%* were also the same. Table 7 better highlights the differences in estimates across different models compared to the *Cr_100%* model. The average of differences in parameters of a model is also provided to give an idea of the difference in the estimated value of parameters across the models. This is calculated as the sum of the % differences of all parameters divided by the number of parameters for each model. The values of average percentage difference also suggest that models trained on samples drawn at the epoch level performed considerably well than those trained on samples drawn at crash and segment levels. Specifically, for *Ep_25%* and *Ep_50%*, the average percentage difference in parameters estimates was about 31% and 34% respectively. These values are considerably smaller than the rest of the models. There was also a considerable improvement in model estimation times for the *Ep_25%* and *Ep_50%* models compared against the *Cr_100%* model. The estimation times were as follows: 159 seconds for *Ep_25%* and 334 seconds for *Ep_50%* and 647 seconds for *Cr_100%*.



a. Change in probability with crash with Speed



b. Change in probability with crash with traffic flow

Figure 3. Change in probability of crashes across different intervals (base = beyond 1 hour of the current epoch).

In summary, the findings from model runs indicate that complete training data is not a necessity to obtain consistent estimates when using the framework. This finding is particularly useful considering that smaller

TABLE 6. Parameter estimates and model fit for samples drawn at various levels.

Variable groups	Variables	Crash level sampling				Epoch level sampling			Segment level sampling		
		Cr_100%	Cr_5%	Cr_25%	Cr_50%	Ep_5%	Ep_25%	Ep_50%	Sg_5%	Sg_25%	Sg_50%
Intercept	Intercept	5.336	7.203	4.586	5.016	5.903	5.619	5.505	7.273	6.321	5.770
Duration dynamics	Time since last crash	-0.254	0.294	0.052	-0.047	-0.261	-0.318	-0.284	0.271	-0.323	-0.039
	Square of time since last crash	0.327	2.550	0.710	0.562	0.493	0.257	0.267	2.538	0.864	0.630
Time of day (base = Night)	Early morning	-0.823	-2.650	-1.078	-0.742	-0.791	-0.824	-0.838	-2.728	-0.521	-1.097
	Late morning	-0.289	-0.605	-0.010	-0.373	-0.601	-0.431	-0.373	-0.637	-0.321	-0.596
	Early afternoon	-1.005	0.221	-0.606	-1.025	-1.208	-1.111	-1.077	0.205	-1.891	-1.206
	Late afternoon	-2.026	-0.894	-2.647	-2.182	-2.465	-2.059	-2.054	-0.892	-2.857	-2.154
	Evening	-0.656	-1.075	-0.065	-0.480	-0.172	-0.574	-0.642	-1.070	-0.307	-0.298
Weather condition (base = Clear)	Cloud rain fog or snow	0.024	-1.470	-0.262	-0.009	0.198	0.082	0.087	-1.536	0.131	-0.194
Lighting condition (base = Daylight)	Dark lighted	0.147	-1.076	0.275	0.251	0.612	0.262	0.224	-1.014	0.251	0.118
	Dark not lighted	0.024	0.179	0.090	-0.002	-0.007*	0.029	0.038	0.178	0.229	-0.065
Terrain (base = Rolling)	Flat	-0.407	-0.482	0.094	-0.154	-0.538	-0.475	-0.378	-0.454	-0.679	-0.040
Land use (base = Mixed)	Rural	-0.376	-0.351	0.222	-0.091	-0.465	-0.372	-0.320	-0.329	-0.907	-0.006*
	Commercial	0.097	0.430	0.052	0.110	0.265	0.109	0.072	0.397	0.448	0.012
Traffic stream characteristics	Average speed	-0.029	-0.023	-0.052	-0.039	-0.020	-0.025	-0.025	-0.019	-0.038	-0.028
	Volume	-0.920	-1.180	-1.366	-1.162	-1.167	-0.901	-0.925	-1.208	-0.845	-1.292
Highway geometry	Number of lanes	0.060	0.405	0.031*	0.055	-0.048*	0.068*	0.022*	0.377	0.419	0.032*
Goodness of fit											
Average initial LL		-0.0831	-0.1169	-0.0776	-0.0812	-0.0798	-0.0791	-0.0853	-0.1179	-0.0558	-0.0710
Average LL at convergence		-0.0139	-0.0163	-0.0132	-0.0137	-0.0126	-0.0131	-0.0134	-0.0166	-0.0093	-0.0122
Average predicted LL		-0.0160	-0.0162	-0.0166	-0.0161	-0.0165	-0.0162	-0.0162	-0.0163	-0.0162	-0.0161
Number of observations in training sample		695,374	24,501	187,330	354,264	34,843	173,843	347,687	24,575	272,146	428,823
Number of observations in the testing sample		68,544									

Note: All parameters are statistically significant at a 5% level of significance except for those accompanied by an asterisk (*).

TABLE 7. Percentage difference in parameter estimates and model fit compared to the model trained on *Cr_100%* data

Variable groups	Variables	Crash level sampling			Epoch level sampling			Segment level sampling		
		Cr_5%	Cr_25%	Cr_50%	Ep_5%	Ep_25%	Ep_50%	Sg_5%	Sg_25%	Sg_50%
Intercept	Intercept	-34.99	14.06	6.00	-10.63	-5.30	-3.17	-36.30	-18.46	-8.13
Duration dynamics	Time since last crash	215.75	120.47	81.50	-2.76	-25.20	-11.81	206.69	-27.17	84.65
	Square of time since last crash	-679.82	-117.13	-71.87	-50.76	21.41	18.35	-676.15	-164.22	-92.66
Time of day (base = Night)	Early morning	-221.99	-30.98	9.84	3.89	-0.12	-1.82	-231.47	36.70	-33.29
	Late morning	-109.34	96.54	-29.07	-107.96	-49.13	-29.07	-120.42	-11.07	-106.23
	Early afternoon	121.99	39.70	-1.99	-20.20	-10.55	-7.16	120.40	-88.16	-20.00
	Late afternoon	55.87	-30.65	-7.70	-21.67	-1.63	-1.38	55.97	-41.02	-6.32
	Evening	-63.87	90.09	26.83	73.78	12.50	2.13	-63.11	53.20	54.57
Weather condition (base = Clear)	Cloud rain fog or snow	6225.00	1191.67	137.50	-725.00	-241.67	-262.50	6500.00	-445.83	908.33
Lighting condition (base = Daylight)	Dark lighted	831.97	-87.07	-70.75	-316.33	-78.23	-52.38	789.80	-70.75	19.73
	Dark not lighted	-645.83	-275.00	108.33	129.17	-20.83	-58.33	-641.67	-854.17	370.83
Terrain (base = Rolling)	Flat	-18.43	123.10	62.16	-32.19	-16.71	7.13	-11.55	-66.83	90.17
Land use (base = Mixed)	Rural	6.65	159.04	75.80	-23.67	1.06	14.89	12.50	-141.22	98.40
	Commercial	-343.30	46.39	-13.40	-173.20	-12.37	25.77	-309.28	-361.86	87.63
Traffic stream characteristics	Average speed	20.69	-79.31	-34.48	31.03	13.79	13.79	34.48	-31.03	3.45
	Volume	-28.26	-48.48	-26.30	-26.85	2.07	-0.54	-31.30	8.15	-40.43
Highway geometry	Number of lanes	-575.00	48.33	8.33	180.00	-13.33	63.33	-528.33	-598.33	46.67
Goodness of fit										
Average percentage change in parameters		599.93	152.82	45.40	113.47	30.94	33.74	609.97	177.54	121.85
Percentage difference in average initial LL		-40.67	6.62	2.29	3.97	4.81	-2.65	-41.88	32.85	14.56
Percentage difference in average LL at convergence		-17.27	5.04	1.44	9.35	5.76	3.60	-19.42	33.09	12.23
Percentage difference in average predicted LL		-1.25	-3.75	-0.62	-3.13	-1.25	-1.25	-1.87	-1.25	-0.62

Note: A negative percentage change in average LL indicates a lower average LL for the model versus the Cr_100% model.

discretizations of inter-crash duration are desired, and the size of expanded data increases considerably for smaller time resolutions. Therefore, when using this framework, modelers can focus their time and resources on using more finer time resolutions or increasing study area than on using larger samples for improving predictive accuracy. Based on findings, we suggest if the data size becomes an issue, crashes should be drawn at the epoch level to obtain consistent estimates. This will reduce computational time considerably with negligible loss in predictive accuracy.

7.2. Effect Of Speed And Volume On Crash Occurrence

As we mentioned earlier, various effects of speed and traffic flow on the occurrence of crashes have been reported in the literature. It is, therefore, necessary to explore the influence of traffic flow parameters on crash occurrence using the current framework. Based on the model estimates, we present the change in the probability of crashes for the range of speed and traffic volumes in Figure 3. Figure 3(a) suggests that for the same speed, the probability of crashes is lower in the second 15-minute interval of an epoch. Similarly, from Figure 3(b) it is evident that for the same traffic volume, crashes are less likely to occur in later intervals of an epoch.

8. Discussion and future implications

This study considered two interstates in the city of Memphis, TN for investigating sampling techniques and sample size requirements for a recently proposed duration model-based crash prediction framework. Despite the relatively smaller study area, the expanded data obtained after reformulation (creation of forecasting epochs) was considerably large. For comparison, the original training data consisted of only 1,174 unique crashes. Upon expansion using a discretization of 1-hour epochs with 15-minute intervals, the number of observations increased to 695,374. Notably, this increase in data size is mainly due to our choice of finer discretization. Nevertheless, the same issues can be expected when applying the framework. The choice of time discretization is crucial in real-time crash prediction modeling since a finer resolution can capture variations in traffic flow conditions more precisely in turn providing meaningful insights on their effects. For example, for a model formulation where the epochs are considerably longer, say, an hour or multiple hours, the effect of speed can be positive implying crashes are less likely to occur in the future. Such a finding would be a considerable departure from those considering shorter epochs such as ours where we report a contradicting result. However, as mentioned earlier, smaller discretization will result in considerable data size and challenge in model estimation. Nevertheless, sampling techniques can be used to reduce the computational complexity and still derive accurate estimates when applying the framework as demonstrated in this study.

Despite challenges with data size, the potential implications of using smaller discretization in the current framework are attractive since real-time data collected at small intervals can better capture the variations in dynamic traffic flow and predict crashes in a segment more accurately. Beyond, crash prediction, applications of this framework can extend to address questions such as ”*How will crashes be temporally distributed in a segment during congestions?”* and “*How likely are crashes to occur during a particular time of a day?*”. Understandably, such an application will require strategic organization of real-time data collection and its processing (data cleaning, expansion, and sampling in this case). This will bring forth considerable challenges concerning data storage and the computational power needed to obtain crash probabilities in real-time. To this end, traffic planners can consider scalable distributed systems where each component can be assigned specific tasks to reduce the computational load and therefore the overall computational time. These systems can be installed and maintained in-house in local or regional Transportation Management Centers where the traffic data for a geographical area is collected and stored. Additionally, data latency (the time taken to transmit the collected data) and bandwidth (amount of data that can be transferred at a time) should also be given due consideration to ensure the data is collected and

analyzed accurately and in real-time. These requirements are inherent to any real-time crash prediction model that includes the current framework.

9. Conclusions

Duration models, which have been widely applied to static risk models for crash prediction have not been employed on dynamic or real-time cash prediction models due to their inability to handle time-varying covariates. A recent study has introduced a duration-based MNL framework that overcomes this issue by dividing the duration between crashes into equal-time intervals and modeling them as alternatives in an MNL wherein the time-varying covariates can be introduced as interval-specific variables. This approach however comes at the expense of a large data size resulting from the discretization of inter-crash duration into equal time intervals to create forecasting epochs. Even for a relatively smaller study area, the expanded data can be expected to be considerably large. To alleviate the computational complexity arising from large data, this study investigated the effect of sampling techniques and sample sizes on the computational time and parameter estimates. In doing so, the assumption of homogenous highway segments and a longer forecasting window adopted in the original study were also addressed. The assumption of homogeneous segments was addressed by segmenting I-40 and I-55 in Memphis, Tennessee based on geometric attributes (number of lanes), terrain (flat or rolling), and flow (posted speed limit). A realistic future window of 1-hour discretized into 15-minute time intervals was used in this study to discretize the duration between crashes. Training the MNL framework on samples drawn using various techniques and its evaluation using testing data revealed that models trained on smaller samples can provide reasonably accurate estimates provided the sampling is done appropriately. Among samples drawn at the crash, epoch, and segment levels, epoch level samples provide the most consistent estimates and better prediction. Results indicate that as little as 25% of samples drawn at the epoch level can reduce training time to 25% of the time needed to train the model using complete data with minimal error in parameter estimates.

Although rigorous and established methods have been adopted in current research, it is not without limitations. Future studies can address three notable limitations in this study. First, the current study ignores unobserved heterogeneity between segments. If unobserved heterogeneity exists, the parameter estimates derived might not be consistent. Second, the current study overlooks traffic movements when segmenting highways. Variation of traffic movements across highway segments can influence crashes. Third, the focus of this study was to explore sampling techniques that make the framework easier to train. Therefore, no comparisons were made with other prediction models. However, a comparison between the current framework and other established real-time crash prediction models in terms of their predictive accuracy is necessary. In particular, ML algorithms have proven to be more accurate than regression techniques. Future studies can compare the predictive abilities of the current framework with machine learning algorithms explored in the literature.

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Author statement

Diwas Thapa: Software, Formal Analysis, Investigation, Writing-Original Draft, Visualization. **Rajesh Paleti**: Conceptualization, Methodology, Software, Writing-Review & Editing. **Sabyasachee Mishra**: Conceptualization, Writing-Review & Editing, Supervision, Project administration.

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